

Siva Akash
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Profiles

LinkedIn

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Programming Languages

C, C++, Embedded C

Microcontrollers & Embedded Platforms

STM32 Series

ESP32

PIC18F Series

NVIDIA Jetson Orin

Communication Protocols

UART

I2C

SPI

CAN

TCP

UDP

Embedded & Core Concepts

ARM Cortex M Architecture

MCU Boot Process and Startup

Bare-Metal Programming

Register-Level Driver Development

Polling and Interrupts Based Designs

OS Concepts

FreeRTOS

Linux Internals

Tools & Softwares

STM32Cube IDE

MPLAB IDE

Proteus

Ki CAD

Git

Summary

Embedded Systems Engineer with 2 years of experience in hardware-software integration. Specializes in designing bare-metal firmware and custom HAL, register-level driver development, seamless hardware-software integration, and delivering highly reliable, fault-tolerant embedded systems.

Experience

Tata Advanced Systems Limited

Dec 2024 - Present

Embedded Software Engineer

- Developed and optimized bare-metal C/C++ firmware for high-reliability R&D and defense sector applications.
- Developed joystick input handling and state machine logic for button and axis events, ensuring precise input processing in real-time embedded environments
- Implemented ICD-compliant UART serial communication protocol over RS-422/RS-485 for interfacing with Various Sensors like MET, Tilt, Encoders, Single Axis Gyroscopes and Multi Axis Gyroscopes, maintaining end-to-end data integrity and validation.
- Developed POST (Power-On Self-Test) and CBIT (Continuous Built-In Test) diagnostic frameworks to monitor sensor health and hardware integrity, ensuring system reliability in mission-critical operating conditions.
- Engineered Control firmware using mathematical calibration algorithms for bore sighting of optical cameras and dynamic gun firing offset compensation, directly contributing to fire control system accuracy.
- Implemented ICD-compliant communication protocol over UDP for LWS Sensor (laser warning subsystem) ensuring real time Laser threat detection and notifying other subsystems.
- Built and integrated multiple custom FPGA PCIe Linux kernel drivers, loading them into the kernel and interfacing with FPGA hardware for precision gun firing pulse generation.
- Implemented Control Logic for DIO peripherals over I2C to strictly manage gun firing sequence generation and essential safety checks.

Projects

Smart PDU

- Designed end-to-end system schematics for a 3-channel Smart PDU, integrating buck regulators, an I2C-controlled DAC for dynamic voltage regulation, and ideal diode controllers for reverse polarity protection.
- Architected register-level firmware and a custom HAL from scratch for an STM32F103C8 MCU, implementing low-level peripheral drivers and an RS-422 command parser to dynamically manipulate converter output.
- Engineered a dual-layer overcurrent protection system, utilizing a hardware analog watchdog for zero-latency cutoff, backed by a non-blocking DMA-driven software loop for continuous ADC current monitoring.
- Developed a custom analog front-end with shunt resistors, current sense amplifiers, and active filtering to deliver precise load metrics to the MCU.
- Maximized system reliability by implementing a fault-tolerant peripheral state machine featuring auto-recovery, watchdog resets, and robust external communication.

Debugging

- ST-Link
- Oscilloscope
- Logic Analyzer
- OpenOCD
- GDB
- Tera Term

Projects

Car Black Box

- Architected bare-metal firmware in C for a PIC18F4580 microcontroller to prototype a vehicle event data recorder, continuously monitoring and logging telemetry such as speed, gear states and collision events.
- Developed custom I2C drivers to interface with RTC for precise event timestamping and an EEPROM for reliable, non-volatile storage for data.
- Integrated a Character LCD (CLCD) driver providing real time display of current vehicle metrics and system states.
- Configured ADC for continuous sampling and hardware external interrupts for low latency, real-time collision detection.
- Implemented a UART communication protocol to enable the extraction of event logs to an external diagnostic console.

Motion Alert System

- Designed and developed a motion detection and alert system using an ESP32 microcontroller and PIR sensor
- Implemented embedded firmware in C for motion sensing, event detection, and real-time alert generation.
- Interfaced PIR sensor and implemented interrupt/polling based logic for motion detection.
- Integrated wireless communication to send instant alerts through a Telegram chatbot for real-time monitoring.

Courses

Emertxe Information Technologies August 2023 - June 2024

Embedded Systems Trainee

Trained extensively in C and bare-metal programming for microcontrollers, Linux internals, and real-time embedded development. Gained hands-on experience with communication protocols including UART, I2C, SPI, CAN. Developed and integrated sensor interfacing modules, implemented real-time data acquisition, and designed embedded system projects

Education

Bharath Institute Of Higher Education And Research June 2019 - May 2023

I have graduated with a Bachelor's degree in Electronics and Communication Engineering, achieving a CGPA of 8.8 out of 10.